Abstract Interpretation (2)

CSE552 Program Analysis — Lecture 19

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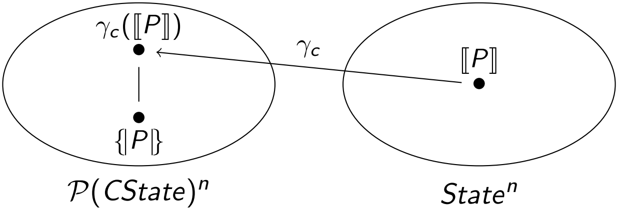
Formal Definition of Soundness

- Let $\gamma : L_2 \to L_1$ be a concretization function where L_1 is the semantic lattice and L_2 is the analysis lattice
- An analysis is sound for a given program P if

$$\{\!|P|\!\} \sqsubseteq \gamma(\llbracket P \rrbracket)$$

  - The concretization of the analysis result over-approximates the semantics of the program
- An analysis is sound if it is sound for every program



Example

```
x = 1;
y = 2;
if input() {
  z = 3;
} else {
  z = y - x;
}
```

$$
\begin{aligned}
&\gamma_c(\llbracket P \rrbracket) \\
&= \gamma_c((\ldots, [x \mapsto +, y \mapsto +, z \mapsto \top])) \\
&= (\ldots, \gamma_b([x \mapsto +, y \mapsto +, z \mapsto \top])) \\
&= (\ldots, \{[x \mapsto d_x, y \mapsto d_y, z \mapsto d_z] \mid \\
&\qquad d_x \in \gamma_a(+) \wedge d_y \in \gamma_a(+) \wedge d_z \in \gamma_a(\top)\}) \\
&= (\ldots, \{[x \mapsto d_x, y \mapsto d_y, z \mapsto d_z] \mid \\
&\qquad d_x \in \mathbb{Z}^+ \wedge d_y \in \mathbb{Z}^+ \wedge d_z \in \mathbb{Z}\}) \\
&\sqsupseteq (\ldots, \{[x \mapsto 1, y \mapsto 2, z \mapsto 3], \\
&\qquad\qquad [x \mapsto 1, y \mapsto 2, z \mapsto 1]\}) = \{\!| P |\!\}
\end{aligned}
$$

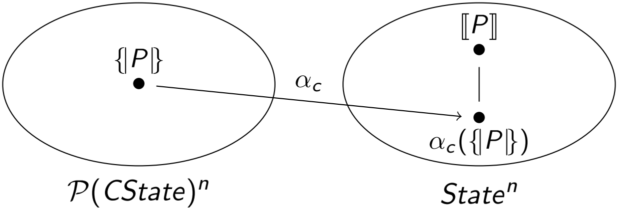
Adjunctions

- Let L_1 and L_2 be complete lattices and $\alpha : L_1 \to L_2$ and $\gamma : L_2 \to L_1$ be functions
- Then, the following is equivalent:
  - α and γ form a Galois connection
  - α and γ are an adjunction between L_1 and L_2:

$$\forall x \in L_1, y \in L_2.\ \alpha(x) \sqsubseteq y \iff x \sqsubseteq \gamma(y)$$

Dual Definition of Soundness

- Let $\alpha : L_1 \to L_2$ be an abstraction function where L_1 is the semantic lattice and L_2 is the analysis lattice
- An analysis is sound for a given program P if
$$\alpha(\{\!|P|\!\}) \sqsubseteq [\![P]\!]$$
  - The analysis result over-approximates the abstraction of the semantics of the program
- An analysis is sound if it is sound for every program



Example

```
x = 1;
y = 2;
if input() {
  z = 3;
} else {
  z = y - x;
}
```

$$
\begin{aligned}
&\alpha_c(\{\!|P|\!\}) \\
&= \alpha_c((\ldots, \{[x \mapsto 1, y \mapsto 2, z \mapsto 3], \\
&\qquad\qquad [x \mapsto 1, y \mapsto 2, z \mapsto 1]\})) \\
&= (\ldots, \alpha_b(\{[x \mapsto 1, y \mapsto 2, z \mapsto 3], \\
&\qquad\qquad [x \mapsto 1, y \mapsto 2, z \mapsto 1]\})) \\
&= (\ldots, [x \mapsto \alpha_a(\{1\}), y \mapsto \alpha_a(\{2\}), z \mapsto \alpha_a(\{3, 1\})]) \\
&= (\ldots, [x \mapsto +, y \mapsto +, z \mapsto +]) \\
&\sqsubseteq (\ldots, [x \mapsto +, y \mapsto +, z \mapsto \top]) = [\![P]\!]
\end{aligned}
$$

Soundness of Abstract Operation

- Let $+ : \mathcal{P}(\mathbb{Z}) \times \mathcal{P}(\mathbb{Z}) \to \mathcal{P}(\mathbb{Z})$
  - $D_1 + D_2 = \{z_1 + z_2 \mid z_1 \in D_1 \wedge z_2 \in D_2\}$
- $\widehat{+}$ is a sound abstraction of $+$ if the following holds:

$$\gamma_a(x_1) + \gamma_a(x_2) \sqsubseteq \gamma_a(x_1 \mathbin{\widehat{+}} x_2)$$

$\widehat{+}$	$\bot$	$-$	0	$+$	$\top$
$\bot$	$\bot$	$\bot$	$\bot$	$\bot$	$\bot$
$-$	$\bot$	$-$	$-$	$\top$	$\top$
0	$\bot$	$-$	0	$+$	$\top$
$+$	$\bot$	$\top$	$+$	$+$	$\top$
$\top$	$\bot$	$\top$	$\top$	$\top$	$\top$

Soundness of $\widehat{+}$ — Example

$$
\begin{aligned}
\gamma_a(+) + \gamma_a(0) &= \{z_1 + z_2 \mid z_1 \in \gamma_a(+) \wedge z_2 \in \gamma_a(0)\} \\
&= \{z_1 + z_2 \mid z_1 \in \mathbb{Z}^+ \wedge z_2 \in \{0\}\} \\
&= \mathbb{Z}^+ \\
&\sqsubseteq \mathbb{Z}^+ \\
&= \gamma_a(+) \\
&= \gamma_a(+ \mathbin{\widehat{+}} 0)
\end{aligned}
$$

Soundness of Abstract Evaluation — Base Cases

eval is a sound abstraction of *ceval* if the following holds:

$$ceval(\gamma_b(\sigma), e) \sqsubseteq \gamma_a(eval(\sigma, e))$$

Prove by structural induction

Case $e = x$:

$$\begin{aligned} ceval(\gamma_b(\sigma), x) &= \{\rho(x) \mid \rho \in \gamma_b(\sigma)\} \\ &\sqsubseteq \gamma_a(\sigma(x)) = \gamma_a(eval(\sigma, x)) \end{aligned}$$

Case $e = n$:

$$ceval(\gamma_b(\sigma), n) \sqsubseteq \{n\} \sqsubseteq \gamma_a(sign(n)) = \gamma_a(eval(\sigma, n))$$

Case $e = \texttt{input}()$:

$$ceval(\gamma_b(\sigma), \texttt{input}()) \sqsubseteq \mathbb{Z} = \gamma_a(\top) = \gamma_a(eval(\sigma, \texttt{input}()))$$

Soundness of Abstract Evaluation — Inductive Case

Case $e = e_1 + e_2$:

- Induction hypothesis:
  - $ceval(\gamma_b(\sigma), e_1) \sqsubseteq \gamma_a(eval(\sigma, e_1))$
  - $ceval(\gamma_b(\sigma), e_2) \sqsubseteq \gamma_a(eval(\sigma, e_2))$

$$
\begin{aligned}
& ceval(\gamma_b(\sigma), e_1 + e_2) \\
& \sqsubseteq ceval(\gamma_b(\sigma), e_1) + ceval(\gamma_b(\sigma), e_2) \\
& \sqsubseteq \gamma_a(eval(\sigma, e_1)) + \gamma_a(eval(\sigma, e_2)) && \text{(induction hypothesis)} \\
& \sqsubseteq \gamma_a(eval(\sigma, e_1) \mathbin{\widehat{+}} eval(\sigma, e_2)) && \text{(soundness of } \widehat{+}\text{)} \\
& = \gamma_a(eval(\sigma, e_1 + e_2))
\end{aligned}
$$

Soundness of *succ*

succ is a sound abstraction of *csucc* if

$$csucc(R, v) \subseteq succ(v)$$

Case $v = \texttt{if}\ x$:

$$csucc(R, v) = T \cup F \subseteq \{true(v), false(v)\} = succ(v)$$

Otherwise:

$$csucc(R, v) = succ(v)$$

Soundness of JOIN

- $CJOIN(v) : \mathcal{P}(CState)^n \to \mathcal{P}(CState)$
- $JOIN(v) : State^n \to State$
- $JOIN$ is a sound abstraction of $CJOIN$ if

$$CJOIN(v)(\gamma_b(\sigma_1), \ldots, \gamma_b(\sigma_n)) \sqsubseteq \gamma_b(JOIN(v)(\sigma_1, \ldots, \sigma_n))$$

$$\begin{aligned}
& CJOIN_v(\gamma_b(\sigma_1), \ldots, \gamma_b(\sigma_n)) \\
& = \{\rho \mid \exists w.\ \rho \in \gamma_b(\sigma_w) \wedge v \in csucc(\rho, w)\} \\
& \sqsubseteq \{\rho \mid \exists w.\ \rho \in \gamma_b(\sigma_w) \wedge v \in succ(w)\} \\
& = \bigsqcup_{w \in pred(v)} \gamma_b(\sigma_w) \\
& \sqsubseteq \gamma_b\left(\bigsqcup_{w \in pred(v)} \sigma_w\right) && \text{(by (1))} \\
& = \gamma_b(JOIN_v(\sigma_1, \ldots, \sigma_n))
\end{aligned}$$

(1) $\gamma_b(\sigma_w) \sqsubseteq \gamma_b\left(\bigsqcup_{w \in pred(v)} \sigma_w\right)$ since $\sigma_w \sqsubseteq \bigsqcup_{w \in pred(v)} \sigma_w$

Soundness of Constraints — Assignment Case

- $cf_v : \mathcal{P}(CState)^n \to \mathcal{P}(CState)$
- $f_v : State^n \to State$
- f_v is a sound abstraction of cf_v if

$$cf_v(\gamma_b(\sigma_1), \ldots, \gamma_b(\sigma_n)) \sqsubseteq \gamma_b(f_v(\sigma_1, \ldots, \sigma_n))$$

$v : x = e$:

- $cf_v(R_1, \ldots, R_n) = \{\rho[x \mapsto z] \mid \rho \in CJOIN(v)(R_1, \ldots, R_n) \wedge z \in ceval(\rho, e)\}$
- $f_v(\sigma_1, \ldots, \sigma_n) = \sigma[x \mapsto eval(\sigma, e)]$ where $\sigma = JOIN(v)(\sigma_1, \ldots, \sigma_n)$

$$\begin{aligned}
& cf_v(\gamma_b(\sigma_1), \ldots, \gamma_b(\sigma_n)) \\
&= \{\rho[x \mapsto z] \mid \rho \in CJOIN(v)(\gamma_b(\vec{\sigma})) \wedge z \in ceval(\rho, e)\} \\
&\sqsubseteq \{\rho[x \mapsto z] \mid \rho \in CJOIN(v)(\gamma_b(\vec{\sigma})) \wedge z \in ceval(CJOIN(v)(\gamma_b(\vec{\sigma})), e)\} \\
&\sqsubseteq \{\rho[x \mapsto z] \mid \rho \in \gamma_b(JOIN(v)(\vec{\sigma})) \wedge z \in ceval(\gamma_b(JOIN(v)(\vec{\sigma})), e)\} && \text{(JOIN)} \\
&\sqsubseteq \{\rho[x \mapsto z] \mid \rho \in \gamma_b(JOIN(v)(\vec{\sigma})) \wedge z \in \gamma_a(eval(JOIN(v)(\vec{\sigma}), e))\} && \text{(eval)} \\
&\sqsubseteq \gamma_b(JOIN(v)(\vec{\sigma})[x \mapsto eval(JOIN(v)(\vec{\sigma}), e)]) \\
&= \gamma_b(f_v(\sigma_1, \ldots, \sigma_n))
\end{aligned}$$

Soundness of Constraints — Entry and Other Nodes

v: entry

$$f_v(\sigma_1, \ldots, \sigma_n) = \top$$

v: other nodes

- $cf_v(R_1, \ldots, R_n) = CJOIN(v)(R_1, \ldots, R_n)$
- $f_v(\sigma_1, \ldots, \sigma_n) = JOIN(v)(\sigma_1, \ldots, \sigma_n)$

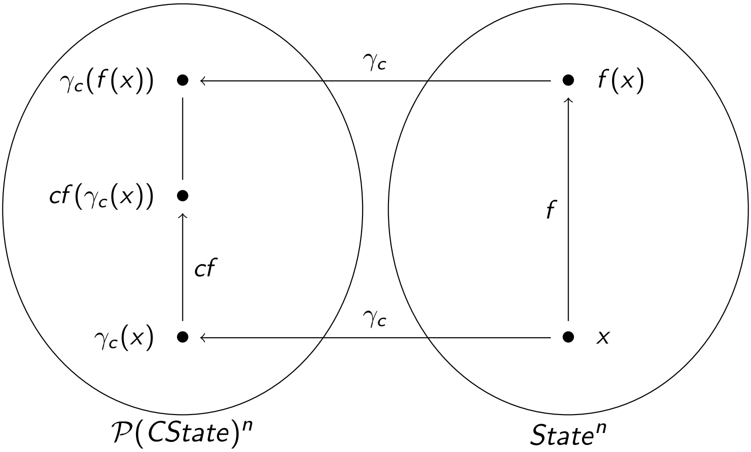
Soundness of Combined Constraints

- $cf : \mathcal{P}(CState)^n \rightarrow \mathcal{P}(CState)^n$
  - $cf((R_1, \ldots, R_n)) = (cf_1(R_1, \ldots, R_n), \ldots, cf_n(R_1, \ldots, R_n))$
- $f : State^n \rightarrow State^n$
  - $f((\sigma_1, \ldots, \sigma_n)) = (f_1(\sigma_1, \ldots, \sigma_n), \ldots, f_n(\sigma_1, \ldots, \sigma_n))$
- f is a sound abstraction of cf if the following holds:

$$cf(\gamma_c(x)) \sqsubseteq \gamma_c(f(x))$$

- Implied by the soundness of each constraint

Soundness of Combined Constraints (cont.)



General Definition of Soundness

- L_1 and L_1' are lattices used by concrete semantics
- L_2 and L_2' are lattices used by abstract semantics
- $\alpha : L_1 \rightarrow L_2$ and $\alpha' : L_1' \rightarrow L_2'$ are abstraction functions
- $\gamma : L_2 \rightarrow L_1$ and $\gamma' : L_2' \rightarrow L_1'$ are concretization functions
- α and γ form a Galois connection, and α' and γ' form a Galois connection
- $cg : L_1 \rightarrow L_1'$ and $g : L_2 \rightarrow L_2'$ are monotone functions

- g is a sound abstraction of cg if

$$cg(\gamma(x)) \sqsubseteq \gamma'(g(x)) \quad \text{for every } x \in L_2$$

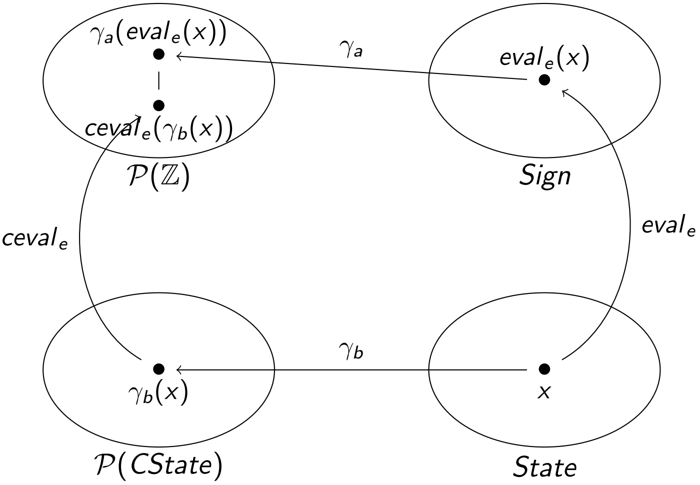
- Or, equivalently, if

$$\alpha'(cg(x)) \sqsubseteq g(\alpha(x)) \quad \text{for every } x \in L_1$$

General Definition of Soundness — Instances

$$
\begin{aligned}
+(\gamma_a^2((x_1, x_2))) &\sqsubseteq \gamma_a(\widehat{+}(x_1, x_2)) \\
ceval_e(\gamma_b(x)) &\sqsubseteq \gamma_a(eval_e(x)) \\
CJOIN_v(\gamma_c((x_1, \ldots, x_n))) &\sqsubseteq \gamma_b(JOIN(v)(x_1, \ldots, x_n)) \\
cf_v(\gamma_c((x_1, \ldots, x_n))) &\sqsubseteq \gamma_b(f_v((x_1, \ldots, x_n))) \\
cf(\gamma_c((x_1, \ldots, x_n))) &\sqsubseteq \gamma_c(f((x_1, \ldots, x_n)))
\end{aligned}
$$

General Definition of Soundness — Diagram



Soundness of Analysis

- $\{\!|P|\!\} = \mathit{lfp}(cf)$
- $[\![P]\!] = \mathit{lfp}(f)$
- The analysis result is sound if

$$\mathit{lfp}(cf) \sqsubseteq \gamma(\mathit{lfp}(f))$$

- Or, equivalently, if

$$\alpha(\mathit{lfp}(cf)) \sqsubseteq \mathit{lfp}(f)$$

Tarski's Fixpoint Theorem

Theorem (Tarski[1])

If L is a complete lattice and $f : L \to L$ is monotone, then f has an lfp, given by

$$lfp(f) = \bigsqcap\{x \in L \mid f(x) \sqsubseteq x\}$$

[1] A lattice-theoretical fixpoint theorem and its applications (Tarski, 1955)

Soundness Theorem

Soundness Theorem

If L_1 and L_2 are complete lattices with monotone functions $\gamma : L_2 \to L_1$, $cf : L_1 \to L_1$, and $f : L_2 \to L_2$ such that $cf \circ \gamma \sqsubseteq \gamma \circ f$, then $lfp(cf) \sqsubseteq \gamma(lfp(f))$

$$\begin{aligned} cf(\gamma(lfp(f))) &\sqsubseteq \gamma(f(lfp(f))) = \gamma(lfp(f)) \\ \gamma(lfp(f)) &\in \{x \mid cf(x) \sqsubseteq x\} \\ lfp(cf) &\sqsubseteq \gamma(lfp(f)) \end{aligned}$$

General Recipe for Proving Soundness

1. Specify the analysis
   - analysis lattice (complete lattice)
   - analysis constraint rules (monotone functions)
2. Specify the collecting semantics
   - semantic lattice (complete lattice)
   - semantic constraint rules (monotone functions)
3. Establish the connection between the semantic lattice and the analysis lattice
   - abstraction function (complete join morphism) or concretization function (complete meet morphism)
4. Show that each constituent of the analysis constraints is a sound abstraction of the corresponding constituent of the semantic constraints
5. Soundness of the analysis follows from the soundness theorem

Widening and Soundness

- When widening is used, $\mathit{lfp}(f) \sqsubseteq [\![P]\!]$
- $\{\!|P|\!\} = \mathit{lfp}(cf) \sqsubseteq \gamma(\mathit{lfp}(f)) \sqsubseteq \gamma([\![P]\!])$
- The same proof strategy applies

Summary

- Soundness can be stated equivalently via $\{\!|P|\!\} \sqsubseteq \gamma([\![P]\!])$ or via $\alpha(\{\!|P|\!\}) \sqsubseteq [\![P]\!]$, linked by the Galois connection
- Analysis soundness decomposes into the soundness of its constituents ($\widehat{+}$, *eval*, *JOIN*, f_v, f), each captured by the general condition $cg \circ \gamma \sqsubseteq \gamma' \circ g$
- By Tarski's theorem, soundness of the constraints lifts to soundness of the analysis: $\mathit{lfp}(cf) \sqsubseteq \gamma(\mathit{lfp}(f))$; the same chain extends to analyses that use widening